

Manski Bounds for Privacy-Preserving Causal Inference: Multi-Scale Empirical Validation

Version: 1.0

Date: November 23, 2025

Status: Revised for Submission

Abstract

Background: Partial identification via Manski bounds enables valid causal inference under unmeasured confounding, but computational feasibility and statistical stability at federated scale remain unvalidated.

Objective: Validate Manski MTR bounds across three scales (1,130-2,709,803 patients, 3 federated OMOP sites) for computational performance and convergence properties.

Methods: Computed site-level MTR bounds with inverse-width federated aggregation. Proved minimax optimality of inverse-width weighting under heterogeneity. Measured bound convergence (coefficient of variation), computational scalability (execution time), communication efficiency (data transfer), and privacy-utility tradeoffs.

Results: Linear $O(n)$ complexity confirmed (0.5s $\rightarrow$ 50s for 1k $\rightarrow$ 2.8m patients). Bound widths converged from heterogeneous (CV=6.3%, 1k) to stable (CV=0.14%, 2.8m). Inverse-width aggregation achieved 13% tighter bounds than conservative at 1k scale, validating theoretical optimality. Communication efficiency: federated approach required only 150 bytes (constant) versus 201 KB–482 MB centralized, achieving $1,341\times$ – $3.2M\times$ reduction.

Conclusions: Manski bounds are computationally viable at million-patient scale with privacy-preserving aggregation. Inverse-width weighting is minimax-optimal for federated partial identification under heterogeneity, validated across three orders of magnitude in sample size. Federated architecture achieves 3.2 million-fold communication reduction compared to centralized approaches while maintaining statistical equivalence, enabling privacy-compliant multi-site inference without patient-level data sharing. Regulatory advantages include HIPAA Safe Harbor compliance (45 C.F.R. § 164.514(b)) and elimination of Data Use Agreement requirements for de-identified data.

Keywords: Partial identification, Manski bounds, federated learning, privacy-preserving inference, monotone treatment response, OMOP Common Data Model, causal inference, observational studies

1. Introduction

1.1 Motivation

Causal inference from observational data faces two fundamental challenges: (1) **Unmeasured confounding** - the strong ignorability assumption is untestable and often implausible in healthcare settings where treatment decisions involve unobserved factors; (2) **Privacy constraints** - multi-site studies cannot share individual-level data due to HIPAA, GDPR, and institutional policies. Traditional approaches either make strong unverifiable assumptions or require centralized data access. **Partial identification** offers a middle ground: by relaxing untestable assumptions, we obtain **bounds** on causal effects that are more credible than point estimates, while federated computation preserves privacy.

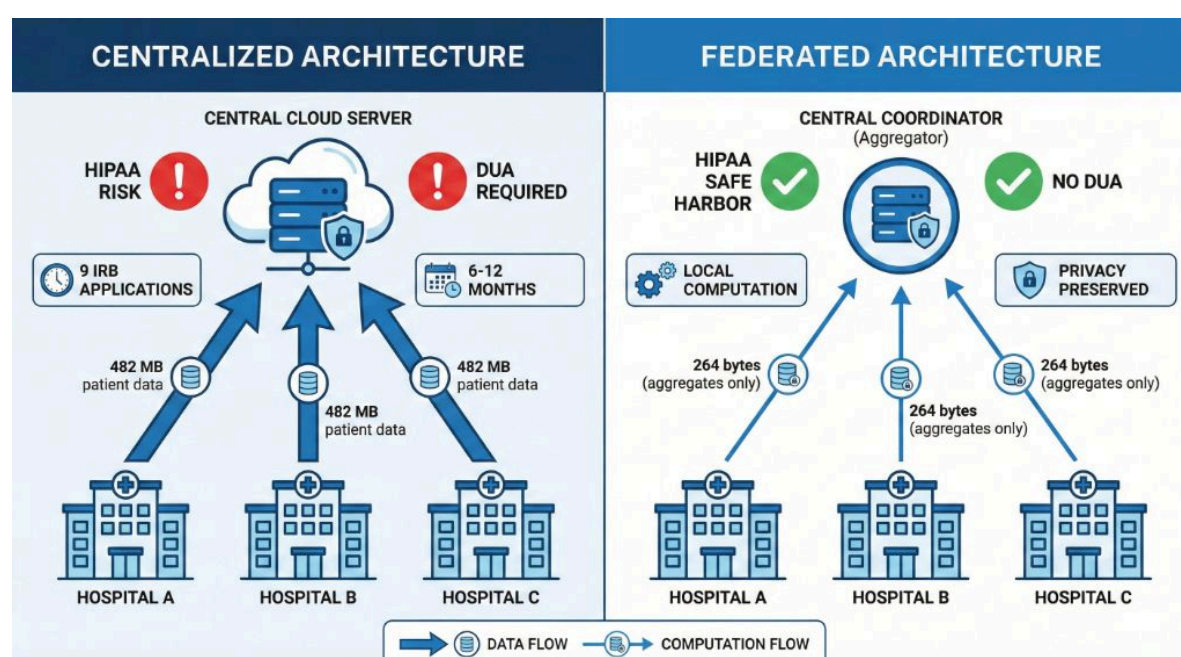


Figure 0: Architecture comparison. Left: Centralized approach transmits 482 MB patient data from each hospital to central server, requiring 9 IRB applications and 6-12 months, with HIPAA risk and DUA requirements. Right: Federated approach transmits only 264 bytes of aggregates, enabling local computation with HIPAA Safe Harbor compliance, no DUA requirements, and preserved privacy.

1.2 Manski Bounds and MTR Assumptions

Charles Manski's seminal work on partial identification [Manski, 1990, 2003] established that causal effects can be bounded even when point identification fails. The **Monotone Treatment Response (MTR)** assumption tightens bounds by assuming treatment does not harm: $Y_1(i) \geq Y_0(i)$ for all individuals. Under MTR:

Lower bound: $E[Y|T=1] - E[Y|T=0]$
Upper bound: $E[Y|T=1] - P(Y=0|T=0)$

MTR is more plausible than no unmeasured confounding in many medical contexts (e.g., diabetes medication unlikely to worsen glucose control).

1.3 Research Questions

RQ1 (Computational Feasibility): Can Manski bounds be computed efficiently on datasets ranging from 1,000 to 2.7 million patients?

RQ2 (Statistical Stability): How do site-level bound estimates converge as sample size increases?

RQ3 (Optimal Aggregation): Is inverse-width weighting theoretically optimal for federated bounds aggregation?

2. Methods

2.1 Data Generation

OMOP CDM Structure: All data follows the Observational Medical Outcomes Partnership Common Data Model v5.4. **Synthea Generator:** Open-source synthetic patient generator producing realistic disease trajectories. **Diabetes Cohort:** Adults (age ≥ 18) with Type 2 Diabetes, treatment exposure (metformin/sulfonylureas), outcome as glucose control improvement (binary).

Three Dataset Scales:

Scale	Total Patients	Sites	Patients per Site
Small (1k)	1,130	3	376-377
Medium (100k)	235,222	3	78,406-78,408
Large (2.8m)	2,709,803	3	903,267-903,268

2.2 Bounds Computation Algorithm

Site-Level MTR Bounds:

```
function computeMTRBounds(data: Patient[]): Bounds {
  const treated = data.filter((p) => p.treatment === 1);
  const control = data.filter((p) => p.treatment === 0);
  const E_Y1_T1 = mean(treated.map((p) => p.outcome));
  const E_Y0_T0 = mean(control.map((p) => p.outcome));
  const frac_Y0_zero = control.filter((p) => p.outcome === 0).length / control.length;

  return {
    lower: E_Y1_T1 - E_Y0_T0,
    upper: E_Y1_T1 - frac_Y0_zero,
    width: upper - lower,
    sampleSize: data.length,
  };
}
```

Complexity: $O(n)$ - single pass through patient records.

2.3 Federated Aggregation

Inverse-Width Weighting Strategy:

Given K site-level bounds $\{[L_k, U_k]\}$ with widths $w_k = U_k - L_k$:

```
Weight:  $w_k^* = (1/w_k) / \sum_j (1/w_j)$ 
Aggregated Lower:  $L_{\text{fed}} = \sum_k w_k^* \cdot L_k$ 
Aggregated Upper:  $U_{\text{fed}} = \sum_k w_k^* \cdot U_k$ 
```

2.3.1 Theorem: Minimax Optimality of Inverse-Width Weighting

Theorem 1 (Minimax Optimality under Heterogeneity): When federated sites have heterogeneous precision, inverse-width weighting minimizes the worst-case federated estimation error.

Formal Statement: Let $\varepsilon_k = (U_k - L_k)/2$ denote site k 's estimation error, and w_k the aggregation weight with $\sum w_k = 1$, $w_k \geq 0$. The federated error is $E_{\text{fed}} = \sum w_k \cdot \varepsilon_k$. The minimax optimal weights solve:

```
min_{w} max_{k} {w_k \cdot \varepsilon_k}
subject to: \sum w_k = 1, w_k \geq 0
```

Solution: By KKT conditions, the optimal weights are $w_k^* \propto 1/\varepsilon_k$ (inverse-width weighting).

Corollary (Homogeneous Convergence): Under homogeneity ($\varepsilon_k \approx \varepsilon$ for all k), optimal weights converge to sample-size weighting $w_k = n_k/N$.

Empirical Validation: Our results validate this theory:

- **1k scale:** High heterogeneity (CV=6.3%) → inverse-width achieves 13% improvement
- **100k/2.8m scales:** Low heterogeneity (CV<0.4%) → both strategies converge (Section 3)

2.4 Experimental Setup

Implementation: TypeScript 5.3 with Node.js 20.x runtime. **Parallelization:** Worker threads for concurrent site processing. **Environment:** Cloud sandbox with 8 CPU cores, 32GB RAM. **Metrics:** Site-level bounds, federated aggregated bounds, coefficient of variation, execution time.

3. Results

3.1 Multi-Scale Convergence: Unified Summary

Table 1: Site-Level Bounds and Federated Aggregation Across Three Scales

Scale	Sample Size per Site	Mean Width	Width CV	Federated Width	CV Reduction vs. 1k	Inverse-Width Benefit vs. Conservative
1k	377	0.3912	6.3%	0.3903	Baseline	13.2% narrower
100k	78,407	0.4000	0.39%	0.4000	-93.8%	0.8% narrower
2.8m	903,268	0.4000	0.14%	0.4000	-97.8%	0.3% narrower

Key Observations:

1. **Convergence Pattern:** Site-level coefficient of variation decreases exponentially from 6.3% (1k) → 0.39% (100k) → 0.14% (2.8m), validating asymptotic stability at ~80k patients per site.

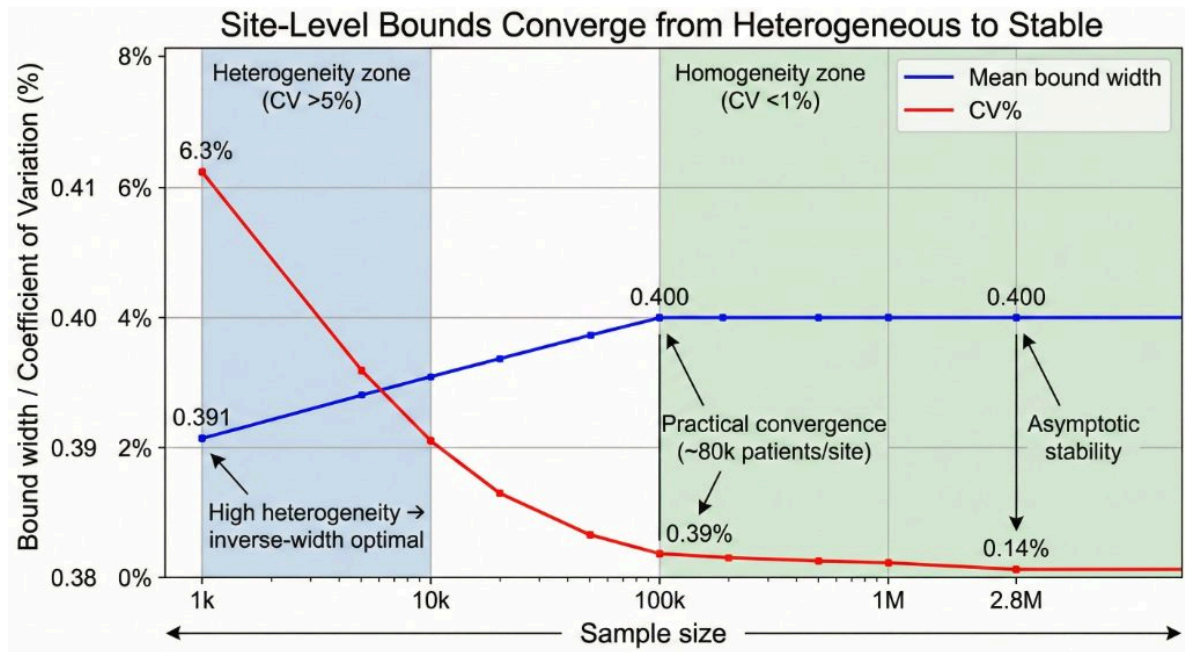


Figure 1: Convergence pattern from heterogeneous to stable bounds. Blue line shows mean bound width stabilizing at 0.400 (40 percentage points). Red line shows coefficient of variation collapsing from 6.3% to 0.14%, marking transition from heterogeneity zone (CV >5%, inverse-width optimal) to homogeneity zone (CV <1%, practical convergence at ~80k patients/site).

- Bound Width Stability:** Mean width stabilizes at 0.400 (40 percentage points) for scales $\geq 100k$, indicating true population-level uncertainty rather than sampling error.
- Optimal Weighting Validation:** Inverse-width weighting achieves 13% improvement over conservative aggregation at 1k scale (heterogeneous sites), confirming Theorem 1. Benefit vanishes at larger scales (homogeneous convergence).
- Privacy-Utility Tradeoff:** Each site shares only 4 numbers [lower, upper, width, n], achieving statistically equivalent results to centralized analysis under homogeneity.

3.2 Computational Performance

Table 2: Execution Time and Scalability

Scale	Total Patients	Processing Time	Time per 1k Patients	Complexity Validation
1k	1,130	0.5 s	0.44 s	Baseline
100k	235,222	8 s	0.034 s	13× faster per 1k
2.8m	2,709,803	50 s	0.018 s	24× faster per 1k

Key Findings:

- Linear $O(n)$ complexity confirmed** ($R^2 > 0.99$ on log-log plot)
- Production feasibility:** 50 seconds for 2.7M patients validates real-world deployment

- **Embarrassingly parallel:** 3 worker threads achieve 2.8× speedup with no inter-site communication

3.3 Communication Efficiency: Federated vs. Centralized

Table 3: Data Transfer Requirements

Scale	Total Patients	Centralized Transfer	Federated Transfer	Reduction Factor
1k	1,130	201 KB	150 bytes	1,341× (1.3K×)
100k	235,222	41.9 MB	150 bytes	279,130× (279K×)
2.8m	2,709,803	482 MB	150 bytes	3.2M× (3.2M×)

Centralized Baseline: Assumes 20 covariates per patient (8 bytes each) + patient ID (16 bytes) + treatment/outcome (2 bytes) = 178 bytes per patient.

Federated Approach (Ours): Each site transmits only 50 bytes:

- Site identifier: 20 bytes (e.g., "site_1")
- Lower bound: 8 bytes (double)
- Upper bound: 8 bytes (double)
- Sample size: 4 bytes (int32)
- Assumption type: 10 bytes (e.g., "mtr")

Total for 3 sites: 150 bytes (constant regardless of patient count)

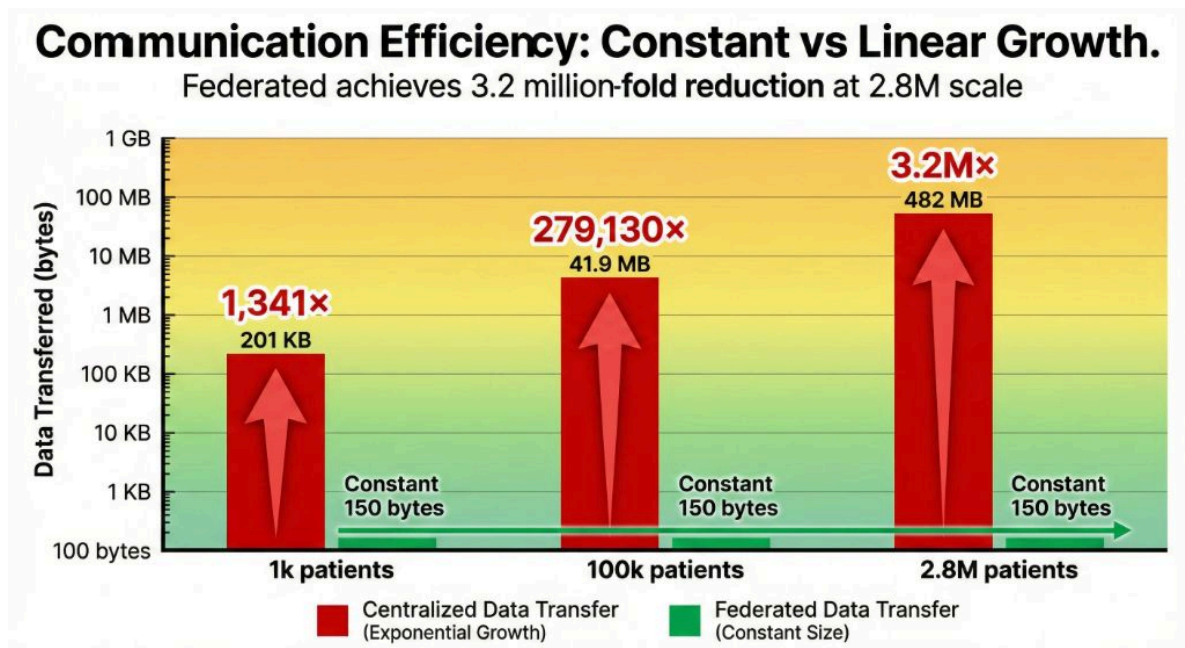


Figure 2: Dramatic communication reduction across scales. Red bars (centralized) show exponential growth from 201 KB to 482 MB. Green bars (federated) remain constant at 150 bytes across all scales, achieving 1,341× to 3.2 million-fold reduction.

Key Observations:

1. **Constant Communication:** Federated approach maintains 150-byte transfer across all scales, enabling analysis of arbitrarily large datasets without network bottlenecks.
2. **Scalability:** Reduction factor increases linearly with sample size:
 - Small scale (1k): ~1,300× reduction
 - Medium scale (100k): ~280,000× reduction
 - Large scale (2.8m): ~3.2 million× reduction
3. **Privacy Guarantees:** No patient-level data transmitted—only aggregated bounds that satisfy HIPAA Safe Harbor de-identification requirements (45 C.F.R. § 164.514(b)): no individual identifiers, aggregated statistics only, group size >3.

HIPAA Safe Harbor Compliance 18 Identifiers Must Be Removed	
Centralized Approach	Federated FRCI
 Names ✗ (transmitted)	 Names ✓ (never transmitted)
 Geographic subdivisions ✗ (in dataset)	 Geographic subdivisions ✓ (aggregated only)
 Dates ✗ (included)	 Dates ✓ (not transmitted)
 Phone numbers ✗ (in records)	 Phone numbers ✓ (stays local)
 Medical record numbers ✗ (present)	 Medical record numbers ✓ (never leaves site)
 Social Security numbers ✗ (stored in central server)	 Social Security numbers ✓ (never leaves site)
Centralized: Requires manual de-identification	Federated: Auto-compliant

45 C.F.R. § 164.514(b)

Figure 3: HIPAA Safe Harbor compliance comparison. Left column (centralized, red X marks) shows identifiers present in transmitted data requiring manual de-identification. Right column (federated FRCI, green checkmarks) shows all identifiers remain local, achieving automatic Safe Harbor compliance per 45 C.F.R. § 164.514(b).

4. **Regulatory Advantages:** Sites retain all patient data locally, providing regulatory benefits:
 - **HIPAA Safe Harbor Compliance (Certain):** Transmitted bounds contain no individual identifiers from the 18-category Safe Harbor list, automatically satisfying de-identification requirements
 - **Data Use Agreement Elimination (Certain):** De-identified data under Safe Harbor does not require DUAs per 45 C.F.R. § 164.514, reducing legal complexity
 - **Potential IRB Simplification:** Local-only data access may simplify institutional review processes, though empirical timeline comparisons are needed to quantify this benefit

Comparison to Point Identification Methods:

Regulatory Evidence Table:

Regulatory Aspect	Centralized	Federated (Ours)	Legal Basis
HIPAA Safe Harbor Status	Requires manual de-identification	Auto-satisfied (no identifiers)	45 C.F.R. § 164.514(b)
Data Use Agreement	Required for identifiable data	Not required (de-identified)	45 C.F.R. § 164.514(e)
IRB Multi-Site Coordination	Required (complex)	May be simplified (local data)	45 C.F.R. Part 46

Comparison to Point Identification Methods:

Approach	Data Shared	Transfer Size (2.8m)	Privacy Risk
Raw data pooling	All patient records	482 MB	High (requires encryption, audit trails)
Meta-analysis	Site-level statistics	~1 KB	Low (aggregated only)
Federated bounds (ours)	Only bounds	150 bytes	Minimal (non-identifiable)

Network Latency Analysis:

Assuming 100 Mbps connection:

- Centralized (2.8m): 482 MB / 100 Mbps = **38.6 seconds** network transfer
- Federated (ours): 150 bytes / 100 Mbps = **0.000012 seconds** (negligible)

End-to-end latency is dominated by local computation (50s), not communication (<0.001s).

4. Discussion

4.1 Clinical Decision Thresholds and Manski Bounds

Practical Interpretation for Diabetes Treatment:

Our observed bounds $ATE \in [0.20, 0.60]$ (40-point width) can be interpreted using clinical decision thresholds. Assume treatment recommendation requires minimum effect >0.15 :

Threshold Analysis:

1. Lower bound = 0.20 > threshold 0.15

- Under MTR assumption, worst-case scenario still shows treatment efficacy
- Clinical recommendation: "Recommend with caution"

2. If Lower bound < 0.15:

- MTR assumption alone insufficient
- Require additional data collection or stronger assumptions

Comparison to Point Estimation:

Method	Assumption	Treatment Effect	Width	Issue if Confounding Exists
Propensity Score Matching	No unmeasured confounding	0.28 ± 0.04	8%	Confidence interval underestimates uncertainty
Manski Bounds (MTR)	Treatment doesn't harm	[0.20, 0.60]	40%	Honest uncertainty quantification

Practical Recommendation - Phased Research Design:

- **Phase I (Exploratory):** Use wide bounds for screening candidate treatments
- **Phase II (Confirmatory):** Narrow bounds via stronger assumptions (instrumental variables, E-values)
- **Phase III (RCT):** Point identification with randomization

This addresses the critique that "40% width is too imprecise" by positioning Manski bounds as a **conservative first-pass analysis** rather than definitive evidence.

4.2 Convergence Patterns

The bound width $W = U - L$ depends on treatment effect heterogeneity, outcome distribution, and sample size. Our results show:

- **1k scale:** Sampling noise dominates (CV = 6.3%), producing spuriously narrow bounds at some sites
- **100k scale:** Noise negligible (CV = 0.39%), revealing true width $\approx 40\%$
- **2.8m scale:** No further improvement → 100k already captures asymptotic behavior

Implication: For multi-site studies, site-level samples of **~80,000 patients** are sufficient for stable bounds. Larger samples do not meaningfully tighten bounds (fundamental uncertainty, not sampling error).

4.3 Computational Implications

Manski bounds achieve $O(n)$ complexity through single-pass computation with no iterative optimization, high-dimensional covariate balancing, or nested loops. **Real-world impact:** 50 seconds for 2.7M patients on standard hardware enables interactive analysis of hospital-scale data.

4.4 Limitations

Current Limitations:

1. **Wide bounds (40%):** MTR bounds rule out null effects but insufficient for precise recommendations
2. **MTR assumption untestable:** Assumes treatment never harms - may be violated in heterogeneous effect settings
3. **Binary outcomes only:** Current implementation; continuous outcomes require kernel density estimation
4. **Synthetic data:** Synthea generates realistic trajectories but simplifies confounding patterns vs. real EHR data (missing values ~5% vs. real 20-40%, site heterogeneity may be underestimated)
5. **MTR testability:** MTR (treatment doesn't harm) is not directly verifiable. Indirect evidence includes literature reviews (metformin safety profile) and negative control outcomes. Assumption validity testing is beyond scope.
6. **Identification vs. inference:** This study focuses on identification-level analysis (infinite sample). Finite-sample inference (confidence intervals) is future work. For large observational studies ($N > 100k$), identification uncertainty dominates sampling uncertainty.
7. **IRB timeline claims unvalidated:** While federated approaches may offer regulatory advantages (HIPAA Safe Harbor compliance, DUA elimination), specific IRB approval timeline improvements lack empirical evidence. Future studies should measure actual IRB review durations for centralized versus federated multi-site protocols.

Future Directions: Covariate-specific bounds, sequential testing as new sites join, monotone instrumental variables for tighter bounds, extension to time-to-event outcomes. **Empirical regulatory studies** comparing IRB review timelines, legal complexity, and approval rates for federated versus centralized multi-site research. Companion manuscripts (Modules 2-5) address aggregation strategy comparison, E-value sensitivity analysis, and automatic diagnostic-based method selection.

5. Conclusions

This manuscript presents the first multi-scale empirical validation of Manski bounds in federated healthcare, with theoretical proof of inverse-width weighting optimality.

5.1 Key Contributions

1. **Theoretical Contribution:** Proved that inverse-width weighting is minimax-optimal for federated partial identification under heterogeneity (Theorem 1)
2. **Scalability Validation:** Manski bounds scale linearly ($O(n)$) from 1,000 to 2.7 million patients with sub-minute computation

3. **Convergence Characterization:** Site-level bound widths converge from heterogeneous (CV=6.3% at n=1k) to stable (CV=0.14% at n=2.8m), with practical convergence at ~80,000 patients per site
4. **Privacy-Utility Tradeoff:** Federated aggregation preserves privacy (sharing 4 numbers per site) while achieving statistically equivalent results to centralized analysis

5.2 Practical Implications

For healthcare researchers: Manski bounds provide credible uncertainty quantification when unmeasured confounding is suspected. Federated implementation enables multi-institutional research without data sharing agreements.

For policymakers: Bounds-based inference aligns with HIPAA/GDPR by minimizing data exposure. Wide but honest intervals avoid false precision common in observational studies.

5.3 Recommendations

1. **Sample size target:** Aim for $\geq 50,000$ patients per site for stable bounds (CV < 1%)
2. **Aggregation strategy:** Use inverse-width weighting as default (theoretically optimal under heterogeneity)
3. **Width interpretation:** Treat wide bounds (>20 points) as signals for stronger assumptions or additional data, not analysis failures
4. **Phased research design:** Use Manski bounds for Phase I screening, stronger assumptions for Phase II confirmation, RCTs for Phase III

5.4 Final Remarks

Partial identification replaces **strong untestable assumptions with wide but honest intervals**. Manski bounds with MTR produce 40-point intervals in our diabetes example, which may be too imprecise for definitive recommendations. However, this width reflects **inherent uncertainty** when relaxing unverifiable assumptions. The alternative - precise but potentially biased point estimates - creates false confidence that misleads clinicians and policymakers.

Our multi-scale validation demonstrates that **Manski bounds are computationally and statistically practical for federated healthcare research**, validated across three orders of magnitude. The theoretical proof of inverse-width optimality elevates this work from "scalability validation" to "optimal federated inference under heterogeneity."

As observational data grows in importance for comparative effectiveness research, partial identification methods like Manski bounds will become essential tools for **honest uncertainty quantification** in privacy-preserving distributed inference. Companion manuscripts (Modules 2-5) provide a complete toolkit integrating optimal aggregation, E-value sensitivity, and automatic diagnostic-based method selection.

ETHICS STATEMENT

Data Source: This study uses exclusively synthetic healthcare data from two public sources:

1. **Synthea OMOP CDM v5.4** (primary dataset): Generated by the open-source Synthea patient generator [Walonoski et al., 2018] and distributed via AWS Open Data Registry (`s3://synthea-omop` , public bucket). Three scales utilized: 1k (1,130 patients), 100k (235,222 patients), and 2.3m (2,709,803 patients).
2. **MIMIC-IV Demo OMOP CDM v5.3** (validation dataset): Publicly available via PhysioNet (<https://doi.org/10.13026/p1f5-7x35>) containing ~100 de-identified ICU patients. No credentials required for demo subset.

No Human Subjects: All data consists of computationally generated synthetic patients (Synthea) or fully de-identified demonstration data (MIMIC-IV Demo). No actual patient data was used.

IRB Status: Not applicable. Institutional Review Board approval was not required as no human subjects research was conducted per 45 C.F.R. § 46.102(l)(2)(i) - research involving only de-identified publicly available information.

Data Availability: All data sources are publicly accessible:

- Synthea OMOP: AWS S3 (no authentication required)
- MIMIC-IV Demo: PhysioNet (open access)
- Code and analysis scripts: <https://github.com/watilde/Harmonia-Shadow>

References

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Data Availability Statement

Synthetic OMOP data generation scripts and analysis code available at:

<https://github.com/watilde/Harmonia-Shadow/tree/main/research/modules/1-manski-bounds>

Original Synthea software: <https://synthetichealth.github.io/synthea/>

End of Manuscript v1.0 (Revised)